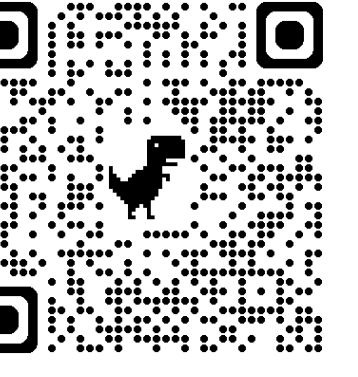


Decoding Pedestrian Crossing Intention from Egocentric Vision via Vision-Language Models



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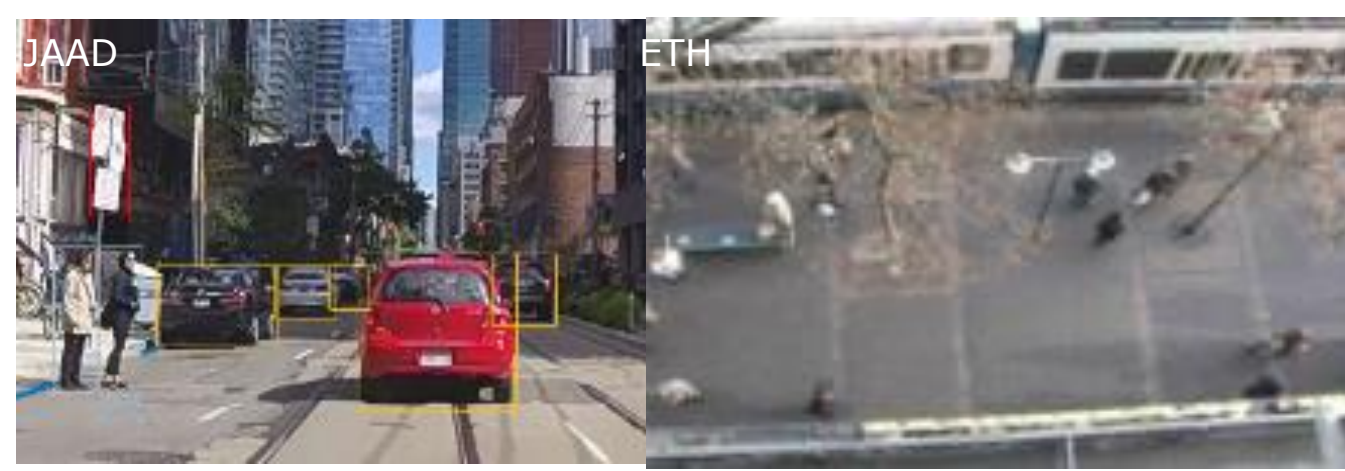
¹ Technical University of Denmark (DTU), ² University of Helsinki, ³ Delft University of Technology



I. Introduction

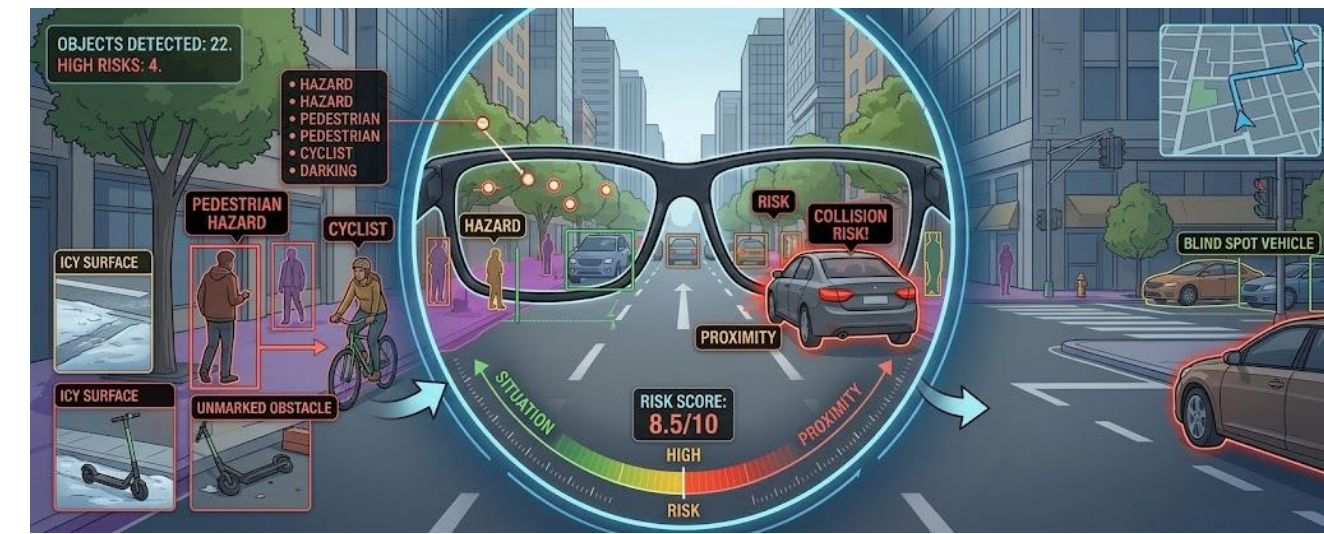
Motivation: The viewpoint gap

Conventionally: Exocentric



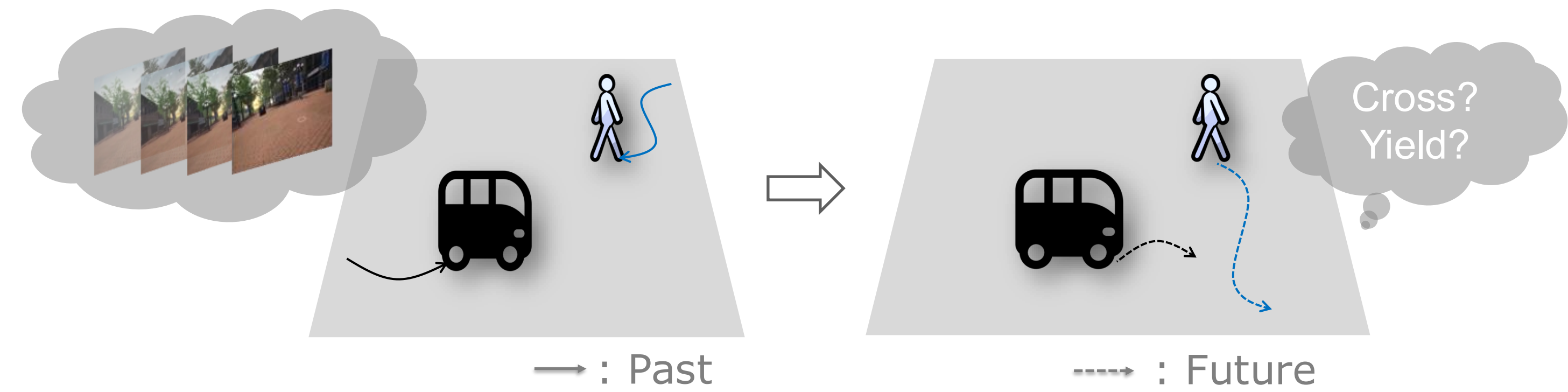
- Fixed or vehicle-mounted cameras
- Provide a stable, global perspective
- Lack true first-person context

Egocentric viewpoint



- Captures direct line-of-sight context
- Underexplored in dynamic urban scenes
- Enables proactive AI assistance, e.g., hazard warnings, accessibility aids

Can we predict pedestrians' future intentions from monocular first-person videos in urban environments?



II. Methodology

STAGE 1: Egocentric Input Data (Virtual Reality derived)

1) Egocentric videos



2) Eye gaze overlay



3) Personal attr.

- Age
- Gender
- ...

STAGE 2: VLM-based Intent Decoding

Video-based Vision Language Models (VLMs)

MLLMs: Qwen, InternVL
VLPs: GroundVQA [1]

Zero-shot via prompting

Text chain-of-thought

Visual prompting

Parameter-efficient fine-tuning

STAGE 3: Output & Justification

Prediction

Cross / Yield

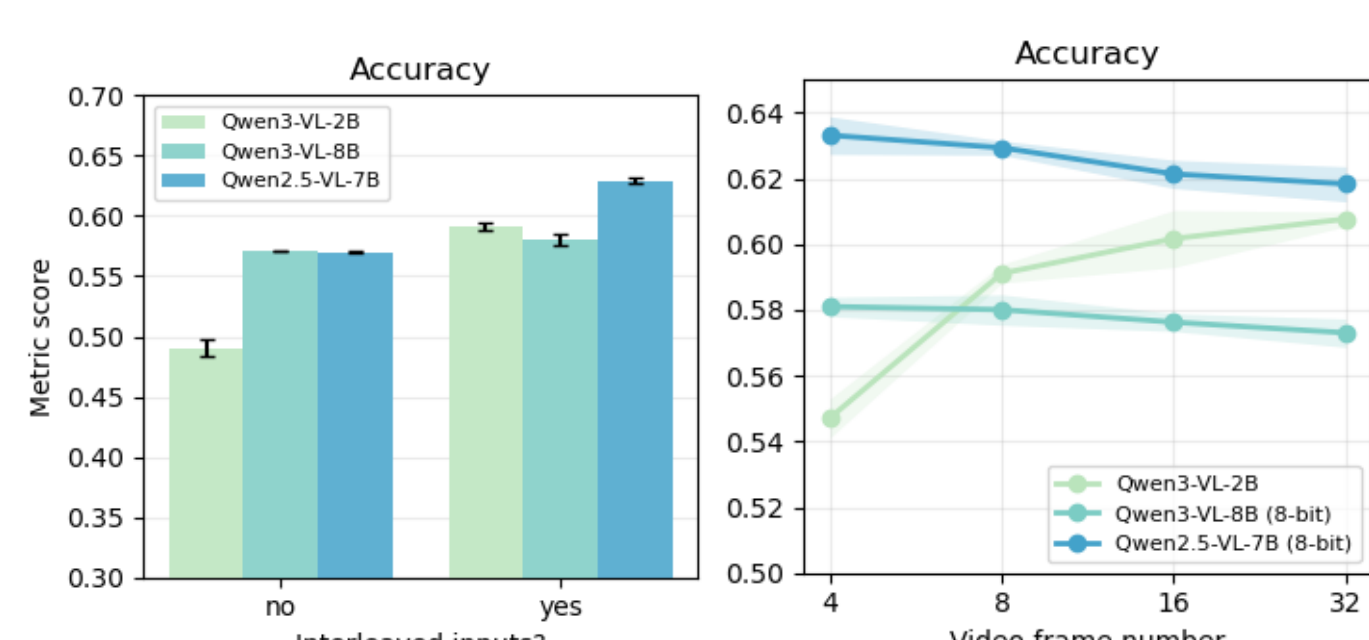
Reasoning & Justification

The video shows a pedestrian walking through an open plaza. There are no visible obstacles or other people directly blocking the path...

III. Results

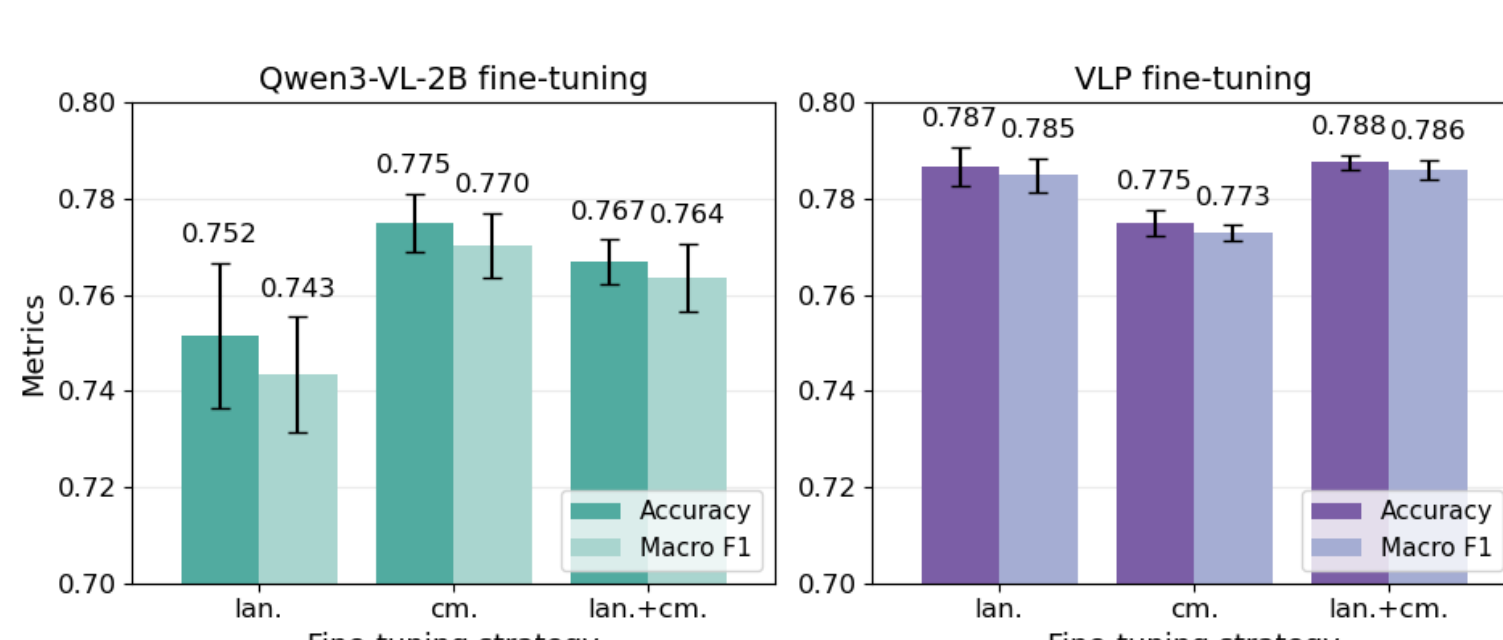
Zero-shot findings:

- Interleaved inputs are effective.

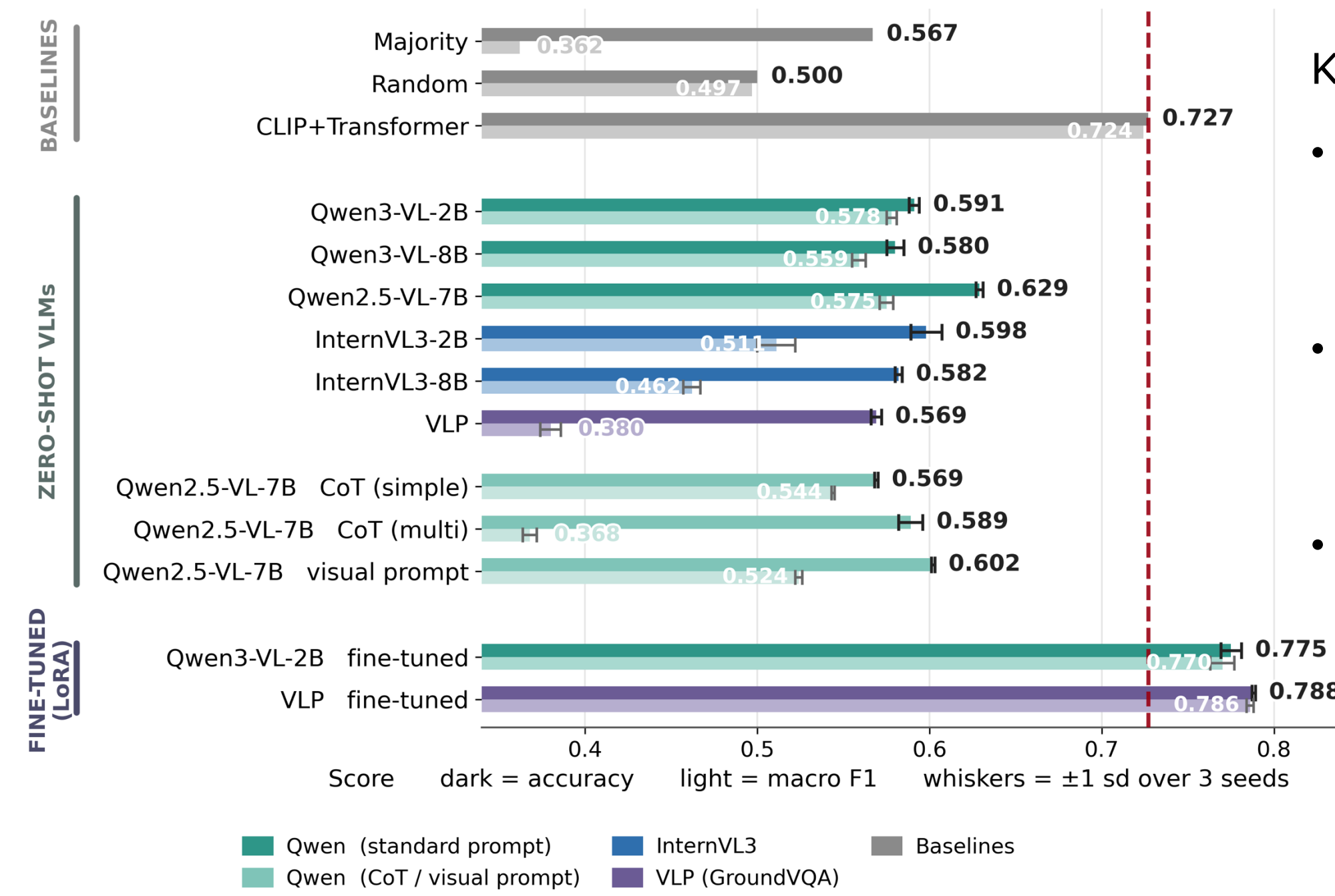


Fine-tuning findings:

- VLP outperforms Qwen, likely due to VLP's egocentric pre-training.



Predicting with Egocentric Video Alone



Key findings:

- Zero-Shot Struggles:** No zero-shot model outperforms CLIP+Transformer baseline.
- Prompting Falls Short** while adding heavy latency (3–5 seconds per sample)
- Fine-Tuning Succeeds**, achieving an 8.4% relative accuracy gain.

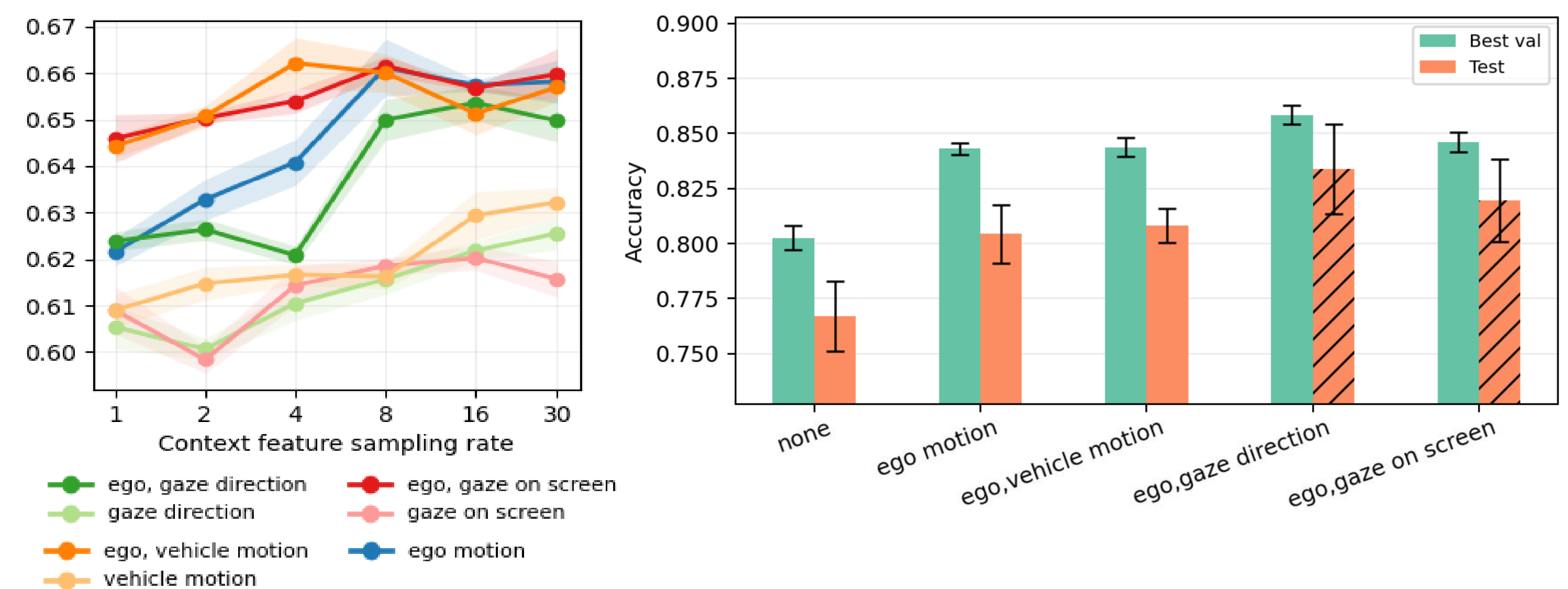
Predicting with Contextual Guidance

Zero-shot findings:

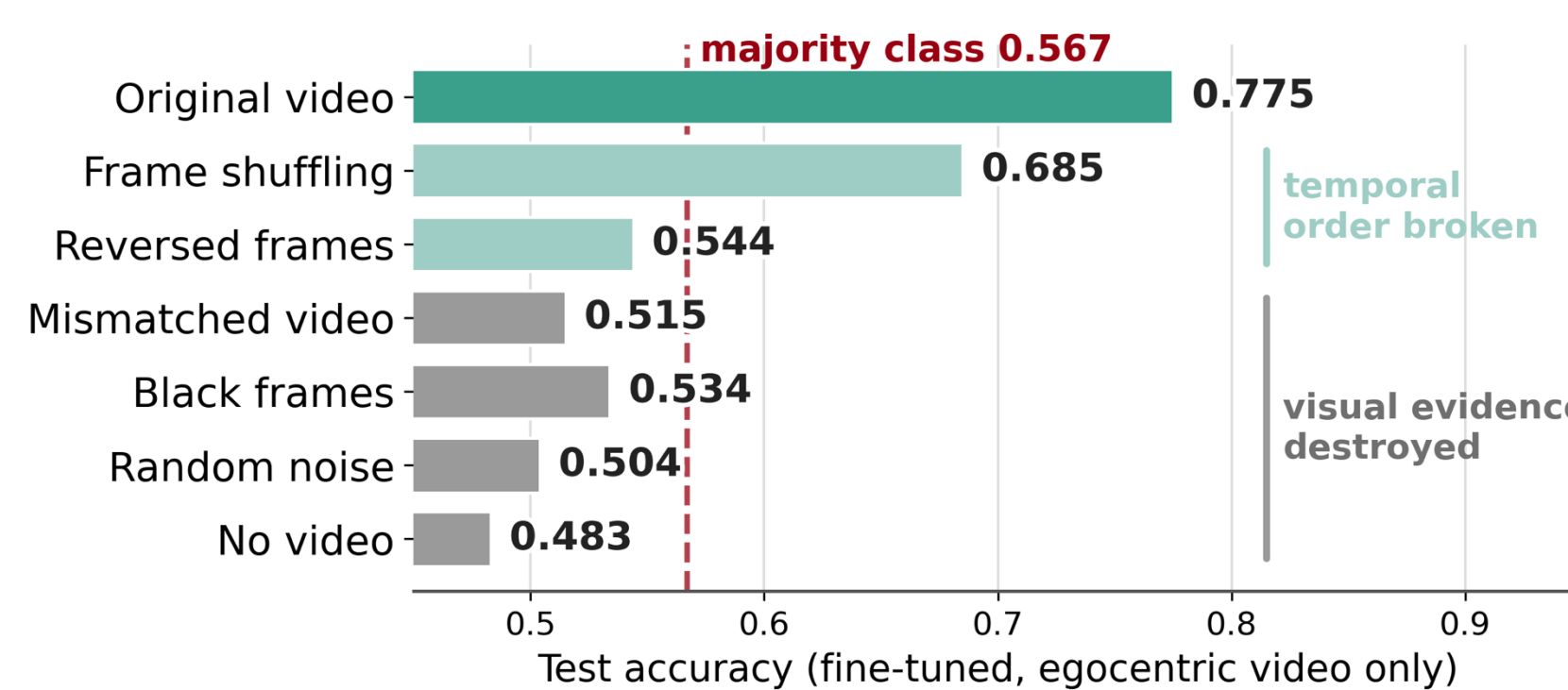
- Gaze Matters:** Gaze-overlaid video consistently outperforms raw video.
- Motion + Context:** Combining both is strictly better than context alone.
- Motion Baseline:** Ego-motion alone serves as a highly informative cue.

Fine-tuning findings:

- Fine-tuning with gaze boosts accuracy by 14.9%**, but can cost general reasoning.

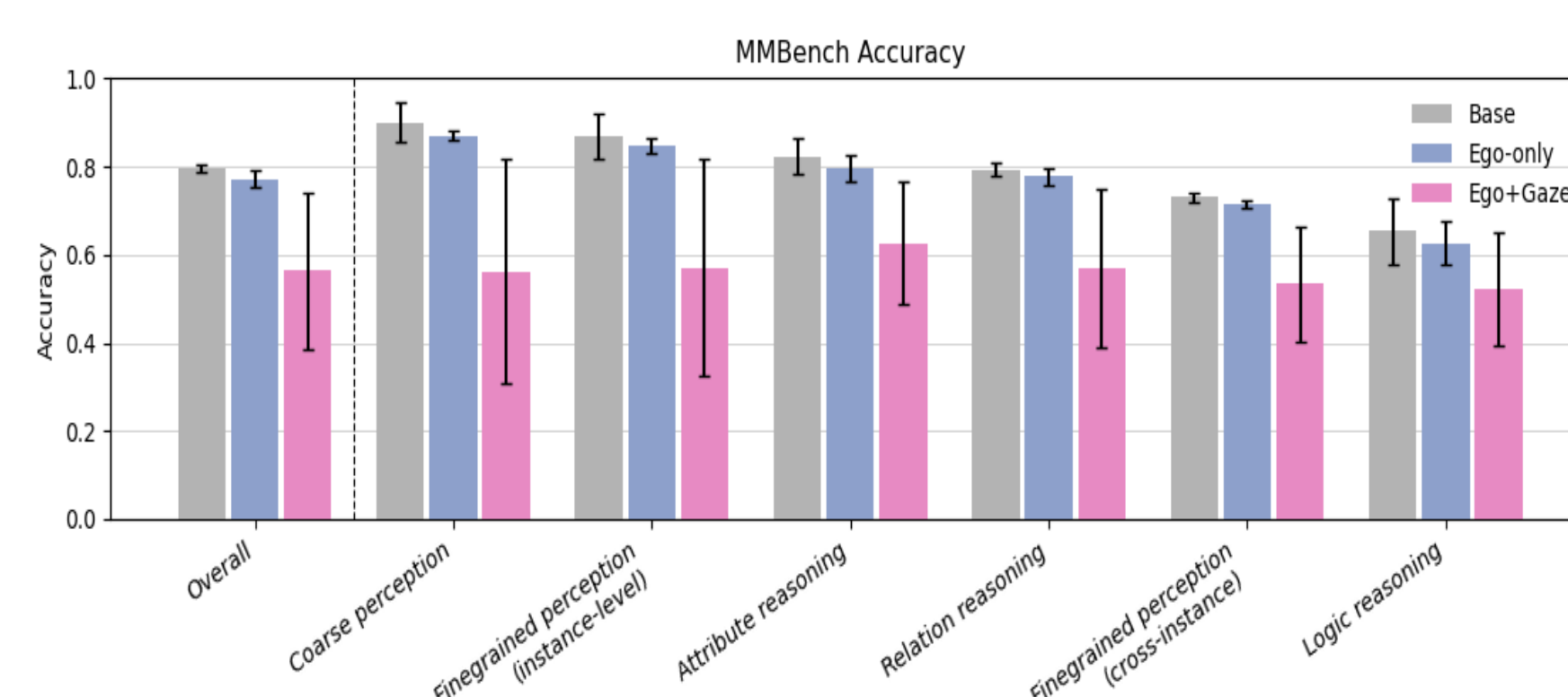


Post-Training Analysis



Key Finding: No Shortcut Learning

- Removing visual evidence drops accuracy to baseline.
- Confirms the model **relies on true visual input**, not language bias.



Impact on General Reasoning

- Ego-only: General reasoning abilities are preserved.
- Ego+Gaze: High variance; performance is highly unstable across seeds.

IV. Conclusion

- Zero-shot VLMs show promise**, outperforming random guessing, but remain **limited in high-level traffic reasoning**.
- Fine-tuning closes it: **gaze plus ego motion** reaches 14.9% above baseline, on real visual and temporal evidence.
- Pedestrian safety need not run through the vehicle:** the same signal can drive assistive eyewear for visually impaired travelers, or a pedestrian-to-vehicle intent.

Limitation: VR-derived data, not yet validated against real-world interactions; on-device operation is a design constraint here.